

fermion scattering can then be rewritten in matrix notation as

$$\begin{aligned}
 S_{\mathbf{p}_1' \sigma_1' n_1' \mathbf{p}_2' \sigma_2' n_2' , \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} &= i(2\pi)^{-2} \delta^4(p_1 + p_2 - p_1' - p_2') \sum_{k'k} \\
 &\times \left[\left(u^\dagger(\mathbf{p}_1' \sigma_1' n_1') \Gamma_{k'} \frac{P(p_{1'} - p_1)}{(p_{1'} - p_1)^2 + M^2 - i\epsilon} \Gamma_k u(\mathbf{p}_1 \sigma_1 n_1) \right) \right. \\
 &\quad \times u_{k'}^*(\mathbf{p}_2' \sigma_2' n_2') u_k(\mathbf{p}_2 \sigma_2 n_2) \\
 &\quad + \left(u^\dagger(\mathbf{p}_1' \sigma_1' n_1') \Gamma_{k'} \frac{P(p_{2'} - p_1)}{(p_{2'} - p_1)^2 + M^2 - i\epsilon} \Gamma_k u(\mathbf{p}_1 \sigma_1 n_1) \right) \\
 &\quad \times u_{k'}^*(\mathbf{p}_2' \sigma_2' n_2') u_k(\mathbf{p}_2 \sigma_2 n_2) \left. \right] \quad (6.3.8)
 \end{aligned}$$

and

$$\begin{aligned}
 S_{\mathbf{p}_1' \sigma_1' n_1' \mathbf{p}_2' \sigma_2' n_2' , \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} &= i(2\pi)^{-2} \delta^4(p_1 + p_2 - p_1' - p_2') \\
 &\times \sum_{k'k} \frac{P_{k'k}(p_{1'} - p_1)}{(p_{1'} - p_1)^2 + m_k^2 - i\epsilon} \\
 &\times \left(u^\dagger(\mathbf{p}_2' \sigma_2' n_2') \Gamma_{k'} u(\mathbf{p}_1' \sigma_1' n_1') \right) \left(u^\dagger(\mathbf{p}_2 \sigma_2 n_2) \Gamma_k u(\mathbf{p}_1 \sigma_1 n_1) \right) \\
 &- [1' \rightleftharpoons 2'] , \quad (6.3.9)
 \end{aligned}$$

where M^2 and m^2 are the diagonal mass matrices of the fermions and bosons in Eqs. (6.3.8) and (6.3.9), respectively. The general rule is that in using matrix notation, one writes coefficient functions, coupling matrices, and propagators in an order dictated by following lines *backwards* from the order indicated by the arrows. In the same notation, the S -matrix for boson-boson scattering in the same theory would be given by a sum of one-loop diagrams, shown in Figure 6.7:

$$\begin{aligned}
 S_{\mathbf{p}_1' \sigma_1' n_1' \mathbf{p}_2' \sigma_2' n_2' , \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} &= -\delta^4(p_1 + p_2 - p_1' - p_2') \\
 &\times \sum_{k_1 k_2 k_1' k_2'} u_{k_1'}^*(\mathbf{p}_1', \sigma_1', n_1') u_{k_2'}^*(\mathbf{p}_2', \sigma_2', n_2') u_{k_1}(\mathbf{p}_1, \sigma_1, n_1) u_{k_2}(\mathbf{p}_2, \sigma_2, n_2) \\
 &\times \int d^4 q \operatorname{Tr} \left\{ \Gamma_{k_2'} \frac{P(q)}{q^2 + M^2 - i\epsilon} \Gamma_{k_1'} \frac{P(q + p_1')}{(q + p_1')^2 + M^2 - i\epsilon} \right. \\
 &\quad \times \Gamma_{k_1} \frac{P(q + p_1' - p_1)}{(q + p_1' - p_1)^2 + M^2 - i\epsilon} \Gamma_{k_2} \frac{P(q - p_2')}{(q - p_2')^2 + M^2 - i\epsilon} \left. \right\} \\
 &+ \dots , \quad (6.3.10)
 \end{aligned}$$

where the ellipsis in the last line indicates terms obtained by permuting bosons 1', 2', and 2. The minus sign at the beginning of the right-hand side is the extra minus sign associated with fermionic loops. Note that after elimination of delta functions there is just one momentum-space